Year 11 Mathematics

Graphs 6

8.1 Using Graphs to Solve Quadratic Equations

Using Graphs to Solve Quadratic Equations

STARTER · 5 MIN

Date: _____ **Title:** Using Graphs to Solve Quadratic Equations

LEARNING OBJECTIVES

- Draw the graph of a quadratic from a table of values
- Solve f(x) = 0 from where the curve crosses the x-axis
- Solve f(x) = k by drawing a horizontal line

STARTER *Answer in your book. No calculators.*

1. Find y when $x = 3$ for $y = x^2$
2. Find y when $x = -2$ for $y = x^2$
3. Find y when $x = -1$ for $y = x^2 - 3x$
4. Solve $x^2 = 25$

Starter — Answers

STARTER

Tick or correct in green pen

1. $3^2 = \mathbf{9}$

2. $(-2)^2 = \mathbf{4}$

3. $1 + 3 = \mathbf{4}$

4. $x = \mathbf{\pm 5}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can draw the graph of a quadratic from a table of values	☐
SILVER	I can solve f(x) = 0 from where the curve crosses the x-axis	☐
GOLD	I can solve f(x) = k by drawing a horizontal line	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Quadratic Graphs

INTRODUCTION · 3 MIN

A quadratic graph is a smooth U-shaped curve. Its graph can be used to solve equations.

Table of values

Substitute each x-value.

$x = -2$: $(-2)^2 = 4$

Smooth curve

Join the points with a smooth curve.

Never join with a ruler

Roots

$f(x) = 0$ where the curve crosses the x-axis.

Read the x-values

f(x) = k

Draw the line $y = k$.

Read where it meets the curve

Examples 1 and 2 — Drawing and Reading Quadratic Graphs

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint roots: where the curve crosses the x-axis

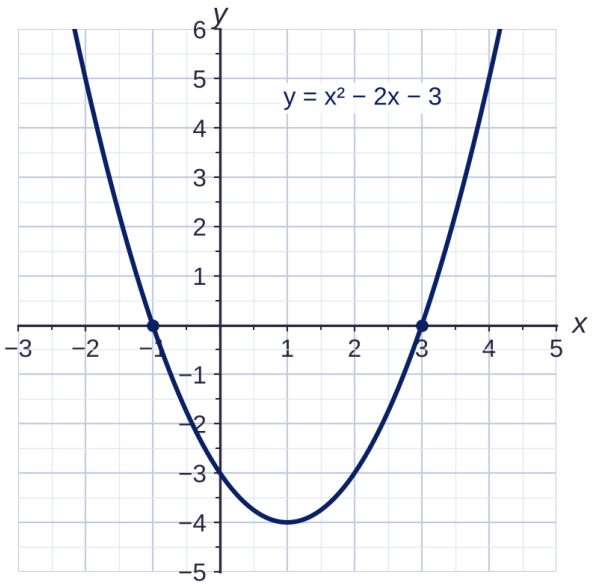
Example 1

Complete the table for $y = x^2 - 2x - 3$.

x	−2	−1	0	1	2	3	4
y		0		−4		0	

Example 2

Use the graph to solve $x^2 - 2x - 3 = 0$.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

x = −2: 4 + 4 − 3 = 5

x = 1: 1 − 2 − 3 = −4

x	−2	−1	0	1	2	3	4
y	5	0	−3	−4	−3	0	5

Tip: put negative x-values in brackets.

Example 2

Where the curve crosses the x-axis

x = −1

x = 3

Tip: read both crossing points.

Task 1

PRACTICE · 9 MIN

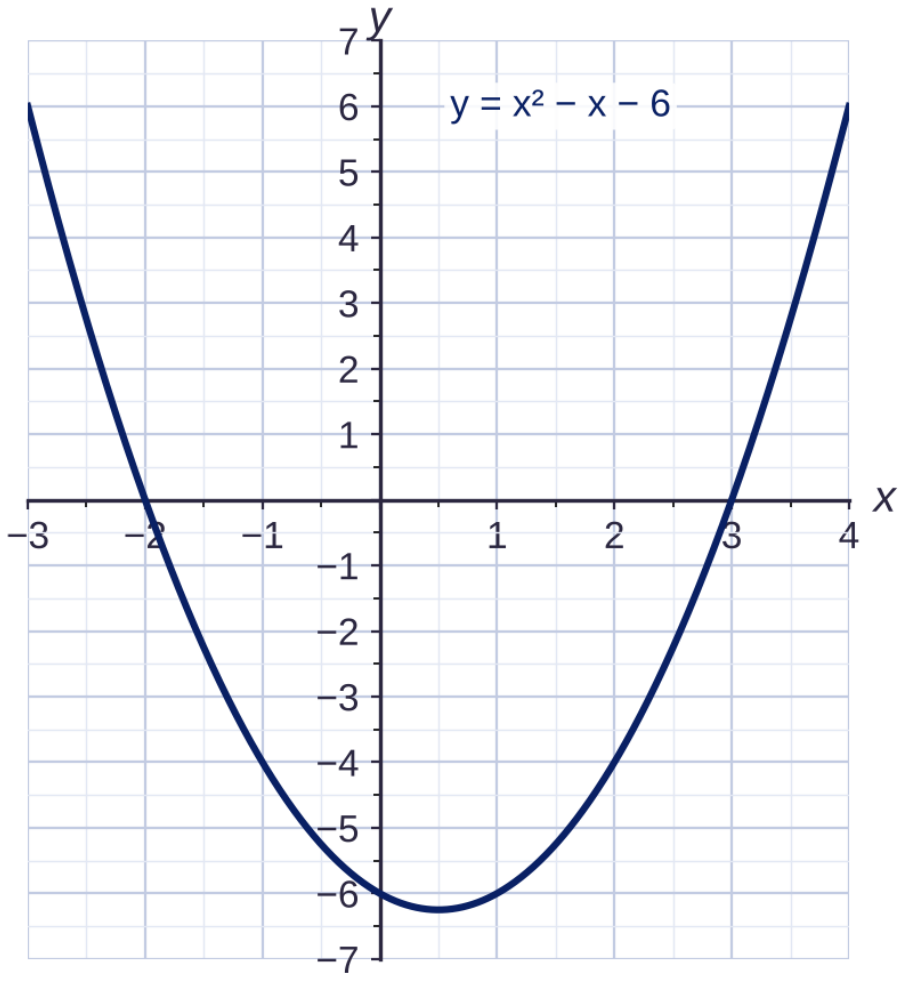
BRONZE

Use the graph of $y = x^2 - x - 6$

Hint roots: where the curve crosses the x-axis

1. Write down the y-intercept
2. Solve $x^2 - x - 6 = 0$
3. Find y when x = 1
4. Write down the turning point
5. Write down the line of symmetry

★ Solve $x^2 - x - 6 = -4$



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **−6**

2. $x = -2$ and $x = \mathbf{3}$

3. $1 - 1 - 6 = \mathbf{-6}$

4. **(0.5, −6.25)**

5. $x = \mathbf{0.5}$

★ Draw $y = -4$, $x = -1$ and $x = \mathbf{2}$

Examples 3 and 4 — Solving f(x) = k

EXAMPLES · 5 MIN

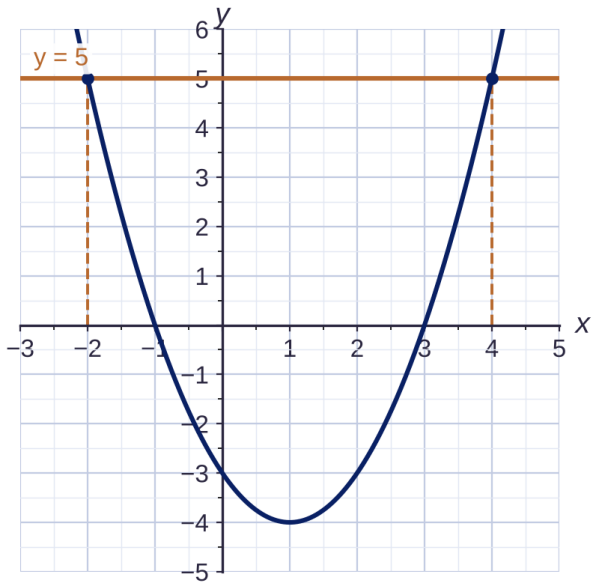
SILVER

Watch and copy into your book

Hint f(x) = k: draw the line y = k

Example 3

Use the graph to solve
$x^2 - 2x - 3 = 5$.



Example 4

Use the graph of
$y = x^2 - 4x + 1$ to solve
$x^2 - 4x + 1 = 3$, to 1 d.p.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Draw the line y = 5

x = −2

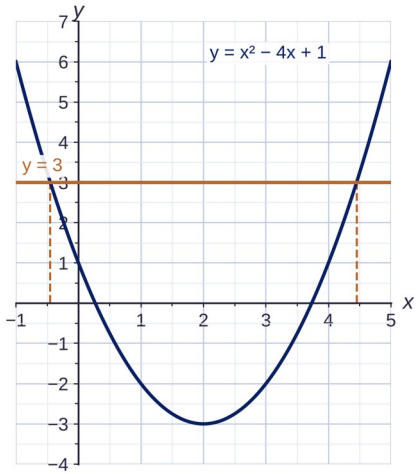
x = 4

Tip: the line meets the curve twice.

Example 4

Draw the line y = 3

x ≈ −0.45 and x ≈ 4.45



Tip: read to the nearest small square.

Task 2

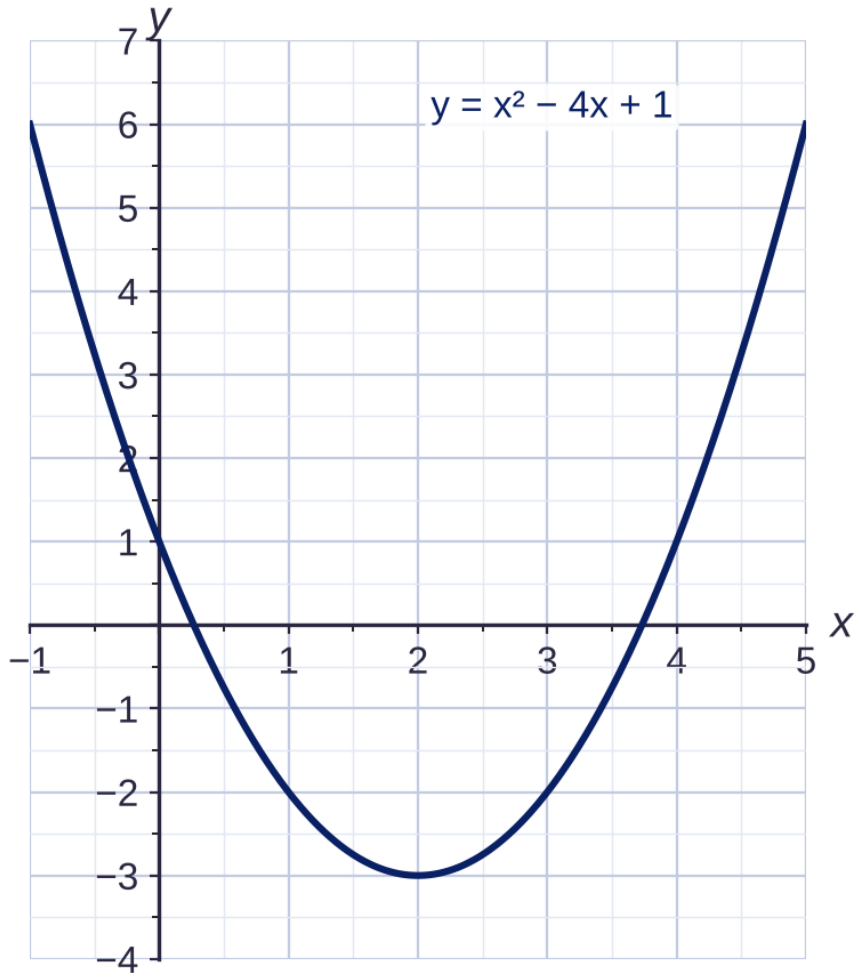
PRACTICE · 9 MIN

SILVER

Use the graph of $y = x^2 - 4x + 1$. Give answers to 1 d.p.

Hint f(x) = k: draw the line y = k

1. Solve $x^2 - 4x + 1 = 0$
2. Solve $x^2 - 4x + 1 = 1$
3. Solve $x^2 - 4x + 1 = -2$
4. Solve $x^2 - 4x + 1 = 4$
5. For what value of k does $x^2 - 4x + 1 = k$ have one solution?



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **$x \approx 0.3$ and $x \approx 3.7$**

2. $x = 0$ and $x = \mathbf{4}$

3. $x = 1$ and $x = \mathbf{3}$

4. **$x \approx -0.6$ and $x \approx 4.6$**

5. Turning point $(2, -3)$, $k = \mathbf{-3}$

Examples 5 and 6 — Rearranging to Use a Graph

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint rearrange so one side is the graph's equation

Example 5

(3 marks)

The graph of $y = x^2 - 4x + 1$ is drawn. Use it to solve $x^2 - 4x - 2 = 0$, to 1 d.p.

Example 6

(2 marks)

Explain why $x^2 - 4x + 1 = -4$ has no solutions.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$x^2 - 4x + 1 = 3$

Draw $y = 3$

$x \approx -0.45$ and $x \approx 4.45$

Tip: add 3 to both sides to match the graph.

Example 6

The minimum value of y is −3

The curve never reaches −4

No solutions

Tip: look at the turning point.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working. Give answers to 1 d.p.

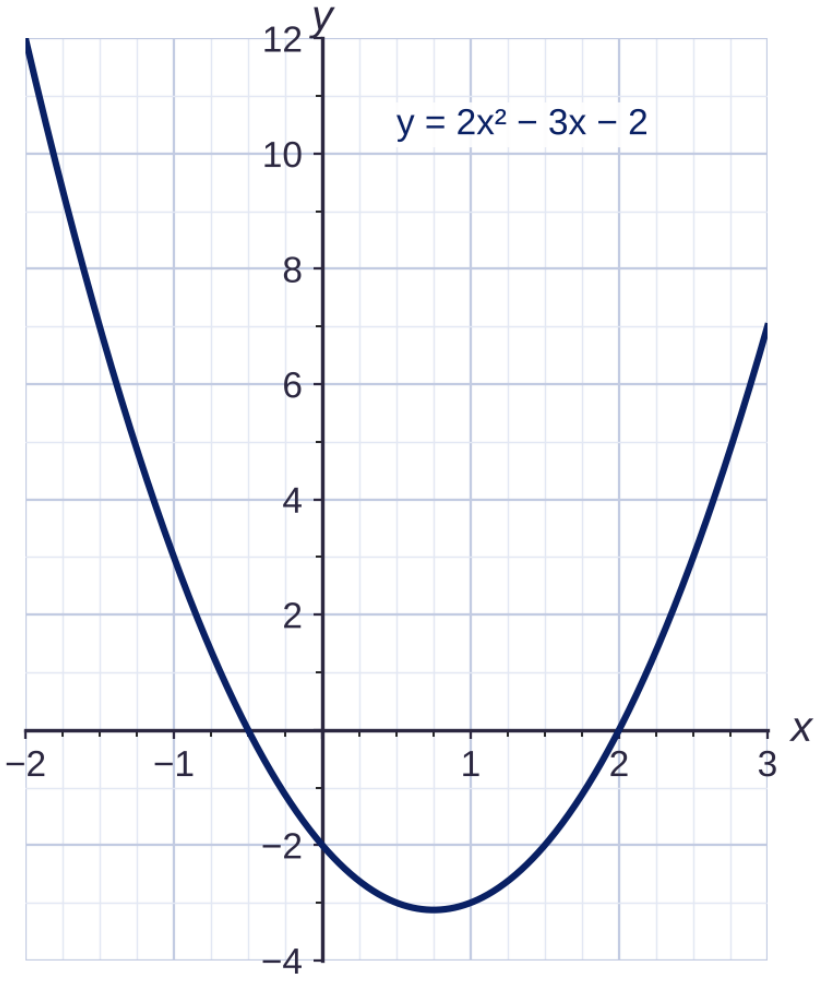
1. Solve $2x^2 - 3x - 2 = 0$ (2 marks)

2. Solve $2x^2 - 3x - 2 = 3$ (2 marks)

3. Solve $2x^2 - 3x = 7$ (3 marks)

4. Estimate the minimum value of y (1 mark)

5. For which values of k does $2x^2 - 3x - 2 = k$ have two solutions? (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $x = -0.5$ and $x = \mathbf{2}$

2. $x = -1$ and $x = \mathbf{2.5}$

3. Draw $y = 5$, $\mathbf{x \approx -1.3}$ **and** $\mathbf{x \approx 2.8}$

4. $\mathbf{y \approx -3.1}$

5. $\mathbf{k > -3.1}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Where does $y = x^2 - 9$ cross the x-axis?

SILVER Which line do you draw to solve $x^2 - 2x - 3 = 5$?

GOLD Using $y = x^2 - 2x - 3$, which line solves $x^2 - 2x = 1$?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x^2 = 9, \quad x = -3$ and $x = \mathbf{3}$

SILVER $y = \mathbf{5}$

GOLD $x^2 - 2x - 3 = -2, \quad y = \mathbf{-2}$

Exam Practice — Reading Quadratic Graphs

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

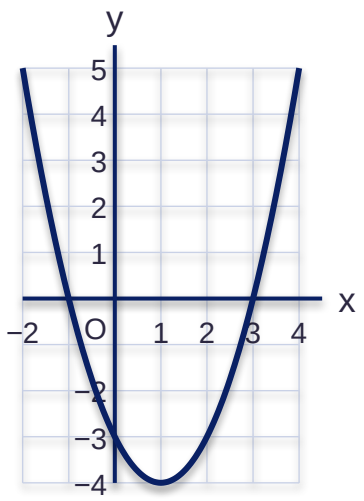
Hint roots: where the curve crosses the x-axis

Question 1

(2 marks)

The graph of $y = x^2 - 2x - 3$ is drawn.

Use it to solve $x^2 - 2x - 3 = 0$.



x = ...

Question 2

(2 marks)

Complete the table of values for

$y = x^2 - 4$

x	−3	−2	−1	0	1	2	3
y	5		−3			0	

Exam Practice — Solving f(x) = k

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint f(x) = k: draw the line y = k

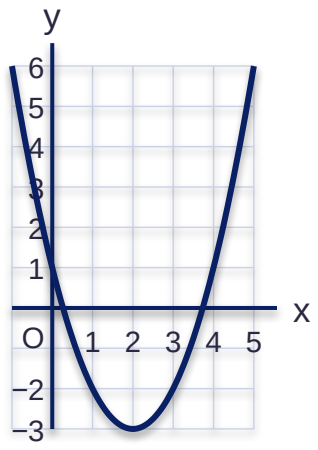
Question 3

(3 marks)

The graph of $y = x^2 - 4x + 1$ is drawn.

Use it to solve $x^2 - 4x + 1 = 2$.

Give your answers to 1 d.p.



x = ...

Question 4

(2 marks)

The graph of $y = x^2 - 2x - 3$ is drawn.

Write down the equation of the line you would draw to solve $x^2 - 2x - 8 = 0$.

y = ...

Exam Practice — Rearranging to Use a Graph

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint rearrange so one side is the graph's equation

Question 5

(2 marks)

The curve $y = x^2 - 4x + 1$ has a minimum point at $(2, -3)$.

For which values of k does $x^2 - 4x + 1 = k$ have no solutions?

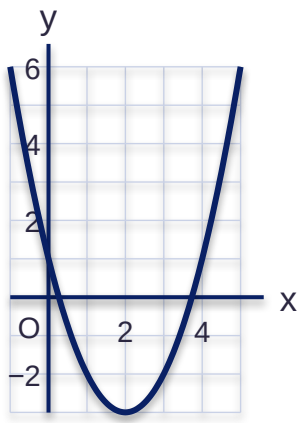
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Question 6

(3 marks)

The graph of $y = x^2 - 4x + 1$ is drawn.

Draw a suitable straight line to solve $x^2 - 5x + 3 = 0$. Give answers to 1 d.p.



x = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

read where the curve crosses — M1

the x-axis

x = −1, x = 3 — A1

Tip: the roots are where y = 0.

Question 2

x = −2: y = 0, x = 0: y = −4 — B1

x = 1: y = −3, x = 3: y = 5 — B1

Tip: $(-2)^2 = 4$, not −4.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

draw the line y = 2 — M1

x = 4.2 — A1

x = −0.2 — A1

Tip: accept 4.1 to 4.3 and −0.3 to −0.1.

Question 4

$x^2 - 2x - 8 = 0$

$x^2 - 2x - 3 = 5$ — M1

y = 5 — A1

Tip: make the left side match the graph.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

y = k must miss the curve — M1

k < −3 — A1

Tip: sketch the line y = k below the curve.

Question 6

$x^2 - 4x + 1 = x - 2$ — M1

draw the line y = x − 2 — M1

x = 0.7, x = 4.3 — A1

Tip: add x − 2 to both sides.